Renormalization group and isomonodromic transformations

S. Srednyak [1]

[1]Duke University

September 26, 2025

Abstract

In this paper, we interpret renormalization group of QED as isomonodromy flow for a pro-finite flat connection on a Riemannian surface with infinitely many punctures. The flat connection is constrained to respect the combinatorics of renormalization algebra of perturbation theory and with internal symmetry of the connection. We also give a local multiparameter verion of this flow, which is important for bound state problems. The renormalization group we consider is exact.

Contents

1 Introduction **1**

2 Regularization **2**

3 Flat bundles and correlation functions **2**

4 Isomonodromy deformations **3**

5 Combinatorics of renormalization **4**

6 Renormalization group flow as a diagonal isomonodromy deformation **4**

7 Local renormalization group **5**

8 Conclusion **5**

1 Introduction

We draw a parallel between isomonodromic deformations and renormalization group in the simplest case of QED. We find that renormalization is equivalnet to isomonodromic flow plus additional symmetry assumptions. We also consider multiparameter version of renormalization, where each generator in the Connes-Kreimer algebra is assigned a separate continous parameter in the renormalization group, and find the analogy of our first conclusion in this case.

There has been much work on various versions of renormalization group in the literature. The geometry of the space of theories is studied as part of Wilsonian approach [1]. Connections to operads is discussed in [2]. In the algebraic field theory, there are discussions in connection with functional renormalization group [3]. Fixed points and c-theorem is discussed [4]. Exact renormalization group is very popular in condensed matter physics [5]. Polchinski's approach and Wetterich refinement is also heavily studied [6]. In the context of gravity physics

there is wide literature on local and functional renormalization group [7]. The conneciton of BV algebras and effective actions there are studied [8]. Despite this huge development, the isomonodromy aspect does not seem to be explained in the literature, even in the simplest QED case.. This is what we do in this paper.

2 Regularization

Def: Dimensional regularization of the integral $\int_\zeta \omega^{d,0}$ of a rational top form $\omega^{d,0}$ over a relative cycle ζ, with boundary containing a component of singularity locus of $\omega^{d,0}$, consists in replacing the Euler $dq_1 \wedge ... \wedge dq_n$ form by $q_0^d dq_1 \wedge ... \wedge dq_n$ with complex d such that the integral over the relative chain becomes convergent.

Before we can use this definition, we compactify the space of internal momenta of the graph.

Def: The graph is called primitive divergent if all of its subgraphs are convergent, while the graph itself is divergent.

Lemma: If the graph is primitive divergent, we can compactify the internal momentum space to a projective space. If the graph is not primitive divergent, each divergent subgraph must be compactified to a projective space.

Th:[?] Each Euler integral $\int_\zeta \omega^{d,0}$ has simple unmovable poles in the regularization parameters.

Each divergent subgraph can be regularized using different regularization parameter ϵ_i. As a result, in the limit we obtain paremeters of the theory, such as running masses or couplings, as functions of infinitely many parametrs $\epsilon_1, ...\epsilon_n, ...$ We exalain in later sections that these paremeters should be assigned to each primitiv divergent graph, to preserve the structure of renormalization algebra.

3 Flat bundles and correlation functions

Renormalization in QED(and other theories) is controlled by a combinatorial structure that is attributed to Bogoliubov and Parasiuk [9] and was studied more recently by Connes and Kreimer [10]. We will call this the BCKP algebra.

It was found more recently that perturbative correlation functions with bare parameters satisfy flatness equations [11]. In the case of propagator, these equations, at each order of perturbation theory, have the form

$$\frac{dI(p,m)}{dp} = \sum \frac{A_i(log(\mu))}{p^2 - r_i m(log(\mu))} I \tag{1}$$

where $p = \sqrt{p^2}$, $I_D(p,m)$ is a vector of master integrals for the diagram $D($ which in general depend on the choice of basis) and r_i are rational numbers depending on the diagram. The position of poles in the denominator is controlled by the bare mass. It can be algorithmically computed from Landau polynomials.

In general, these diagram functions must be regularized, and the connection matrices $A_i(log(\mu))$ depend only on the regularization parameter and not on the momentum or the bare mass.

The differential equation for the photon propagator takes identical form with different connection matrices A_i. We work in a particular gauge.

We now turn to the discussion of differential equations for the vertex function. We define Master Integrals(MIs) as

$$I_\omega(p_1, p_2, m_a) = \int (dq) q^\omega 1/\prod D_i^{n_i} \ . \tag{2}$$

These MIs form a basis of a vector bundle defined on the space (p_1, p_2, m_a). Integration By Parts identities (IBPs) with homogeneity conditions in D_i lead to an exact Cayley-Koszul complex

$$\cdots \to I_n \to I_{n+1} \to \cdots$$

where $I_n = \oplus_{|\omega|=n} I_\omega$. This complex staturates if n is chosen large enough. As a result, we can solve for the MIs in the following form

$$I_n = AI_{n-1} + BI_{n-2} \tag{3}$$

for some matrices A, B. In more detail, these equations can be written as

$$\frac{\partial I_\omega(p_1,p_2,m_a)}{\partial s} = \sum_{R,\omega'} \frac{A_{s,\omega,\omega',R}(p_1,p_2,m_a)}{\prod_{I\in R} L_I^{n_I}} I_{\omega'} \tag{4}$$

for some explicit polynomials $L_I(p_1,p_2,m_a)$ with integer coefficients.

These equations constitute an example of general flatness equations in multiple dimension of the form

$$\frac{\partial f(x_\mu)}{\partial x_\mu} = \Omega_\mu(x_\mu) f = \sum \frac{A_{I,\mu}}{\prod_{i\in I} L_i} f \tag{5}$$

for polynomial L_I and polynomial matrices $A_{I,\mu}$. Consistency conditions take the form

$$\frac{\partial}{\partial x_\mu}\Omega_\nu - \frac{\partial}{\partial x_\nu}\Omega_\mu = [\Omega_\mu, \Omega_\nu] \ . \tag{6}$$

This system of equations constitutes a stringent restriction on the polynomial matrices $A_{I,\mu}$. For example, in the case

$$\Omega_\mu = \frac{A_{I,\mu}}{L_I} + \frac{A_{I,J,\mu}}{L_I L_J} \tag{7}$$

we must have

$$[A_{I_1,I_2,\mu}, A_{I_3,I_4,\nu}] = \sum_{s=1\ldots 4} L_{I_s} C_{s,I_1,\ldots,I_4,\mu\nu}, \ I_p \neq I_q \tag{8}$$

for some polynomial matrices $C_{s,I_1,\ldots,I_4,\mu\nu}$ that satisfy further consistency conditions

$$C_{4,I_1,I_2,I_3,J,\mu\nu} + C_{4,I_1,I_2,I_3,K,\mu\nu}, \ J \neq K \ . \tag{9}$$

4 Isomonodromy deformations

The flatness equations from the previous section must be consistent with the renormalization group flow. In the case of the equation

$$\frac{df}{dt} = \sum \frac{A_i}{t - t_i} f \tag{10}$$

the condition of the isomonodromy takes the following Schlesinger form

$$\frac{\partial A_i}{\partial t_j} = \frac{[A_i, A_j]}{t_i - t_j} \tag{11}$$

$$\frac{\partial A_i}{\partial t_i} = \sum_j \frac{[A_i, A_j]}{t_i - t_j} \tag{12}$$

These equations can be specialized to the case when $t_i = r_i x$, where r_i are fixed numbers and x is a single varying parameter.

We discuss now isomonodromy deformations of flat bundles where base space has dimension larger than 1

$$\frac{\partial f}{\partial x_\mu} = \Omega_\mu(x) f \tag{13}$$

where the flat connection is algebraic. If the connection Ω_μ is multidimensional *Fuchsian*, then

$$\Omega_\mu(x) == \sum \frac{A_{I,\mu}(x)}{\prod_{i\in I} L_i(x)} \ . \tag{14}$$

Def: *Consistency manifold is the stratified space of algebraic matrices $A_{I,\mu}(x)$ that satisfies flatness condition identically.*

This stratified space $C(L_{I_1}, ..., L_{I_n})$ depends on the point $L_{I,\omega}$ in the space of polynomials $L_{I,\omega}$.

Remark: In non perturbative QFT it is necessary to consider $L_I(x)$ that are non polynomial, e.g., entire (in the sense of Vojta). In this case, we are dealing with consistency in the space of series.

Our task now is to formulate conditions for isomondromy. The isomonodromy condition foliates the manifold $C(L_{I_1}, ..., L_{I_n})$ into leaves that correspond to the same monodromy. We write the monodromy matrices as

$$M(l) = Texp(\int_l \Omega_\mu(x)dx^\mu) \tag{15}$$

where l is a closed loop in the space x representing an element in $\pi_1(\mathbb{C}^n - \cup L_I)$.

$$M(l) = Texp(\int \Omega_\mu(x)dx^\mu) = \sum M_n(l) \tag{16}$$

$$M_n(l) = \int \Omega(x_1)dx_1 \int^{x_1} \Omega(x_2)dx_2 .. \int^{x_{n-1}} \Omega(x_n)dx_n \tag{17}$$

5 Combinatorics of renormalization

Renormalization relies on representation of correlation functions as iterated integrals. There is an alternative approach to correlation functions. In perturbation theory, correlation functions satisfy an overdetermined system of differential equations (i.e. they form solution sheaf of a holonomic D-module). This D-module is equivalent to a flat connection after we expand the space of basis integrals. Instead of fixing renormalization point, we can use the boundary conditions at infinity in the kinematic space. Therefore, the transcendental constants of perturbation theory [**?**] are encoded in motivic geometry of the complement of the singularities of the correlation function [**?**, **?**, **?**].

Divergent parts in perturbation theory are isolated in a small number of topological types of diagrams. For example, in QED these are the photon and electron propagators, and the vertex. Renormalization procedure consists of identification of divergent parts of a diagram and regularizing the divergent part. We will use dimensional regularization throught the paper.

6 Renormalization group flow as a diagonal isomonodromy deformation

We argue that renormalization group flow for the coupling, mass and field renormalization is equivalent to the isomonodromy flow for the flat bundle equations together with BCKP symmetry for the connection.

The field renormalization factor is computed as the residue on mass shell

$$S(p^2) \approx \frac{iZ_2(log(\mu))}{\hat{p} - m(log(\mu))} \tag{18}$$

Resummation of regularized series takes the form

$$S(p) = \frac{i}{\hat{p} - m(\log(\mu)) - \Sigma(p, m(\log(\mu)))} \tag{19}$$

Term by term in perturbation theory, $S(p, \alpha(log(\mu)), m(log(\mu)))$ is a flat bundle in p with regular singularities at $r_i m(\log(\mu))$. We can consider the renormalization group flow on the coefficients for the full resummed connection

$$\frac{\partial I(p)}{\partial p} = \sum_i \frac{A_i(\log(\mu))}{p - r_i m(log(\mu))} \tag{20}$$

where for simplicity we passed to the variable $p = \sqrt{p^2}$. It is important to note that $m(log(\mu))$ here is the full renormalized mass flow, so it also depends on the coupling flow $\alpha(log(\mu))$. $A_i(log(\mu))$ depend on the renormalized

coupling and $log(\mu)$. This equation is actually a new equation compared to the previous ones. The previous ones had only finite set of singularities and the bundle was finite dimensional. This equation is actually infinite dimensional. The number of singularities is also infinite. So we are dealing with isomonodromy problem for a profinite connection. Now since this connection already includes the resummation of divergences, it contains the renormalization group flow for both $m(\log(\mu))$ and for the matrices $A_i(log(\mu))$. The matrices $A_i(log(\mu))$ are very symmetric. This is the BCKP symmetry that we referred to above. This equation must be consistent with the exact flow equation. Therefore, we conclude that it is consistent under the joint isomonodromy for all the coefficients $m_i = r_i m(log(\mu))$. This is a special direction in the space of possible flows of a more general equation

$$\frac{\partial J(p)}{\partial p} = \sum_i \frac{A_i}{p - m_i} J(p) \tag{21}$$

We therefore identify our 1-parameter flow with a subflow in this larger space.

7 Local renormalization group

We pass now to the discussion of extension of renormalization procedure to multiparameter renormalization. In parametric integrals, there are multiple infinite families of graphs associated with the generators of the CK algebra. We propose to associate a separate renormalization parameter $log(\mu_a)$ wiht each of these families. Then the removal of singularities still holds, because the combinatorics of counterterms is encapsulated exactly by the CK algebra, as demonstrated in the papers [**?**]. In this verion of renormalization, the mass becomes a function of multiple scales $m(log(\mu_1), ..., log(\mu_n), ...)$ and correspondingly for the coupling and the scale factors. The flatness condition is still similar, of the form

$$\frac{\partial I(p)}{\partial p} = \sum_i \frac{A_i(\log(\mu), ..., log(\mu_n), ...)}{p - r_i m(log(\mu_1), ..., log(\mu_n), ...)} \tag{22}$$

with the exception that the poles of the connection depend on the family of parameters $\mu_1, ..., \mu_n, ...$, and correspondingly for the connection coefficients. The discussion on the consistensy of these equations with the generalized flow in multiparameter RG μ_i still holds. The space of RG parameters is embedded into the full space of movable parameters m_i in a way consistent with combinatorics of the BCKP algebra.

8 Conclusion

We believe that the second version of renormalization group is the right version as it incapsulates more fully the combinatorics of BCKP algebra. This fact should be compared with the combinatorics of other spaces accociated with flat connections, such as Grothendieck-Teichmuller space, see the discussion in [**?**].

We note that the multiparameter version of renormalization group what we discussed can be considered as the perturbative version of local renormalization group [12]. In this version, the space of paremeters has a distinct combinatorial structure, which we believe is important for applications such as renormalization in bound state problems [13](which was an important motivation for this work). The issue of fixing all these scales μ_i self consistently is of central importance for renormalization in bound state problems, such as multiloop Lamb shift computation, and to first principle calculation of the proton mass. We will consider these issues in future publications.

References

[1] Kevin Costello. *Renormalization and effective field theory.* Number 170. American Mathematical Soc., 2011.

[2] Jean-Louis Loday and Nikolay M Nikolov. Operadic construction of the renormalization group. In *Lie Theory and Its Applications in Physics: IX International Workshop*, pages 191–211. Springer, 2013.

[3] Edoardo D'Angelo, Nicolo Drago, Nicola Pinamonti, and Kasia Rejzner. An algebraic qft approach to the wetterich equation on lorentzian manifolds. In *Annales Henri Poincare*, volume 25, pages 2295–2352. Springer, 2024.

[4] Zohar Komargodski and Adam Schwimmer. On renormalization group flows in four dimensions. *Journal of High Energy Physics*, 2011(12):1–20, 2011.

[5] Oliver J Rosten. Fundamentals of the exact renormalization group. *Physics Reports*, 511(4):177–272, 2012.

[6] Juergen Berges, Nikolaos Tetradis, and Christof Wetterich. Non-perturbative renormalization flow in quantum field theory and statistical physics. *Physics Reports*, 363(4-6):223–386, 2002.

[7] Pedro F Machado and Frank Saueressig. Renormalization group flow of f (r) gravity. *Physical Review D—Particles, Fields, Gravitation, and Cosmology*, 77(12):124045, 2008.

[8] Alberto S Cattaneo, Pavel Mnev, and Nicolai Reshetikhin. Classical bv theories on manifolds with boundary. *Communications in Mathematical Physics*, 332:535–603, 2014.

[9] NN Bogoliubov and OS Parasiuk. On the subtraction formalism in multiplicating of causal singular functions. *Izv. AN SSSR, Ser. Mat*, 20(5):585–610, 1956.

[10] Alain Connes and Dirk Kreimer. Hopf algebras, renormalization and noncommutative geometry. In *Quantum field theory: perspective and prospective*, pages 59–109. Springer, 1999.

[11] Stefano Laporta. High-precision calculation of multiloop feynman integrals by difference equations. *International Journal of Modern Physics A*, 15(32):5087–5159, 2000.

[12] Alessandro Codello, Roberto Percacci, and Christoph Rahmede. Investigating the ultraviolet properties of gravity with a wilsonian renormalization group equation. *Annals of Physics*, 324(2):414–469, 2009.

[13] Krzysztof Pachucki and Ulrich D Jentschura. Two-loop bethe-logarithm correction in hydrogenlike atoms. *Physical review letters*, 91(11):113005, 2003.